FunCoup 6: advancing functional association networks across species with directed links and improved user experience

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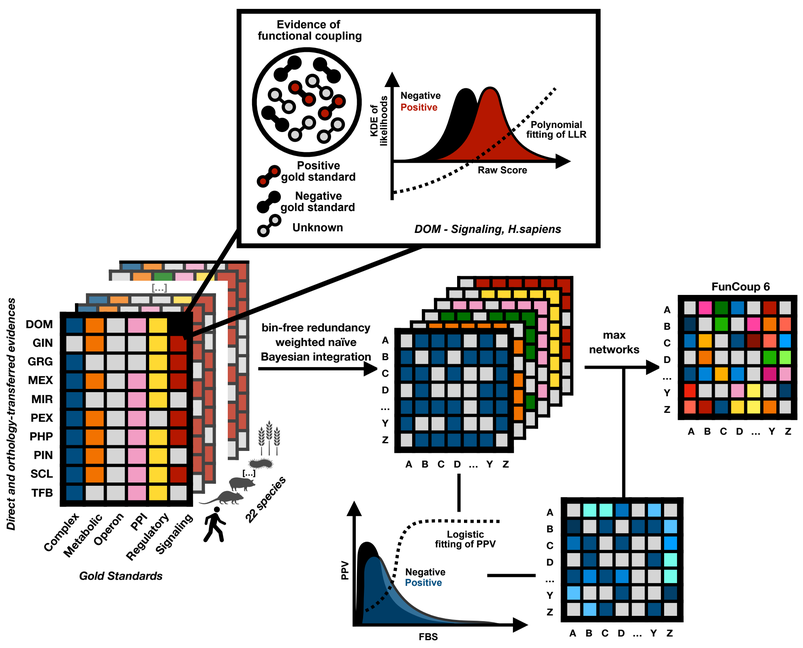
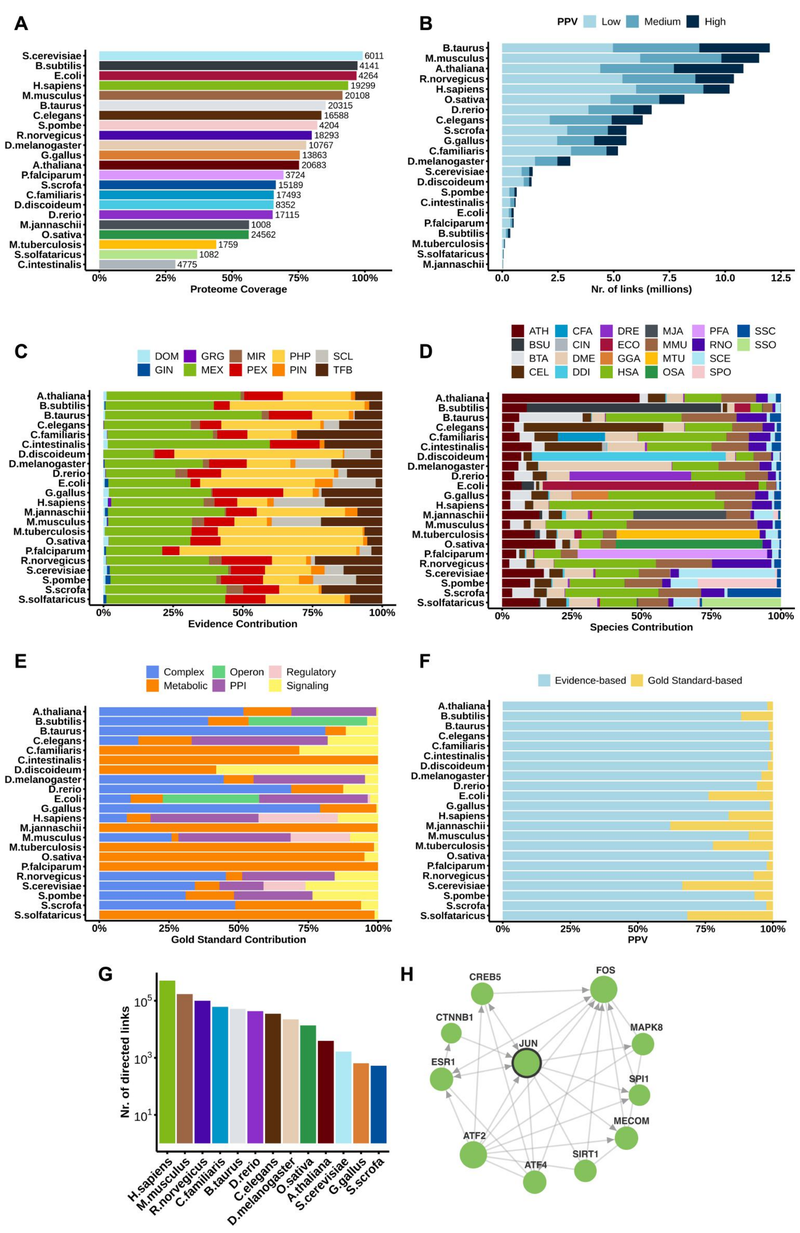
Background

Biological cells function as an integrated whole, forming a complex network of interactions. Thus, mapping gene interactions is essential for understanding cellular mechanisms. FunCoup integrates a wide range of high-throughput interaction data, both direct physical and indirect ones, into functional association networks to better understand the roles of all genes in an organism.

New bin-free network training workflow

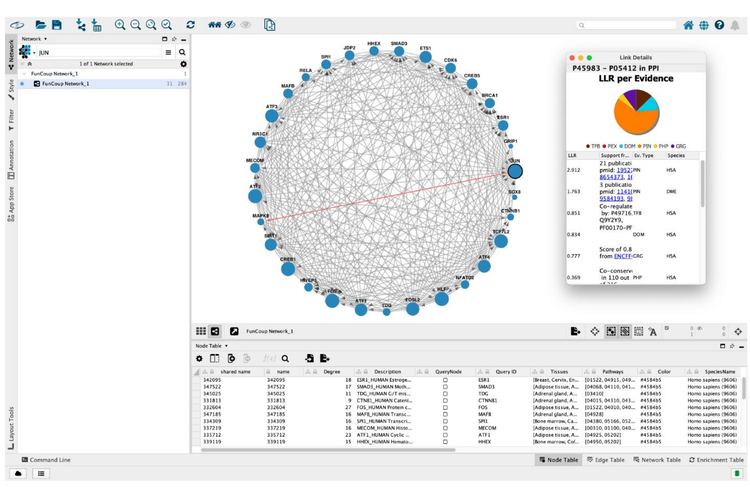


Evidence-based networks for 22 species



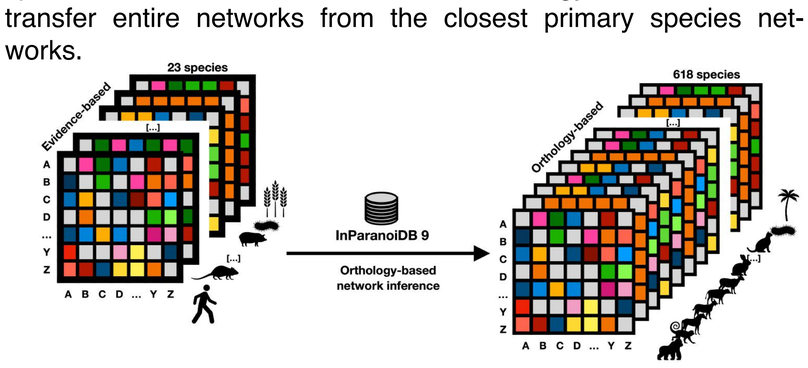
New API functionalities and the Cytoscape App

The FunCoup Cytoscape App lets users query the FunCoup 6 API with gene IDs to get interaction networks in JSON format. Users can customize species and expansion algorithms through an options panel. The app converts these networks into Cytoscape-compatible structures for detailed analysis.

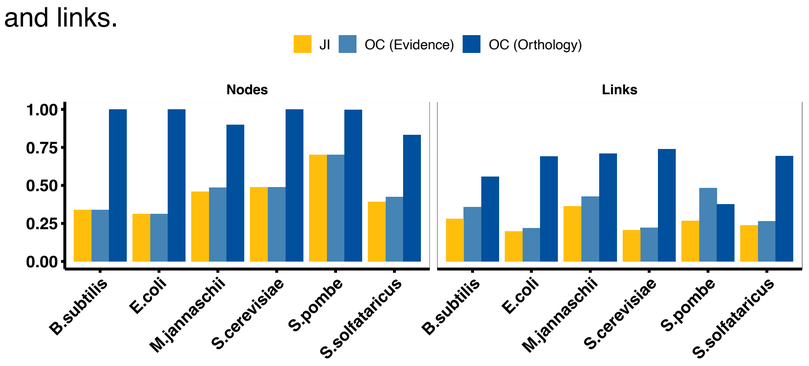


Orthology-transferred networks for 618 species

In FunCoup 6, networks were generated for all remaining 618 species in InParanoidDB 9 via a new orthology-based method to transfer entire networks from the closest primary species networks.

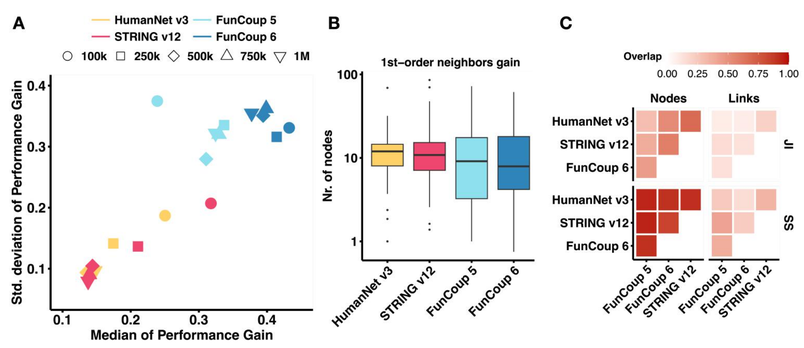


We compared FunCoup evidence-based (Evidence) and orthology-transferred (Orthology) networks for six species in terms of Jaccard index (JI) and overlap coefficient (OC) similarity of nodes and links.



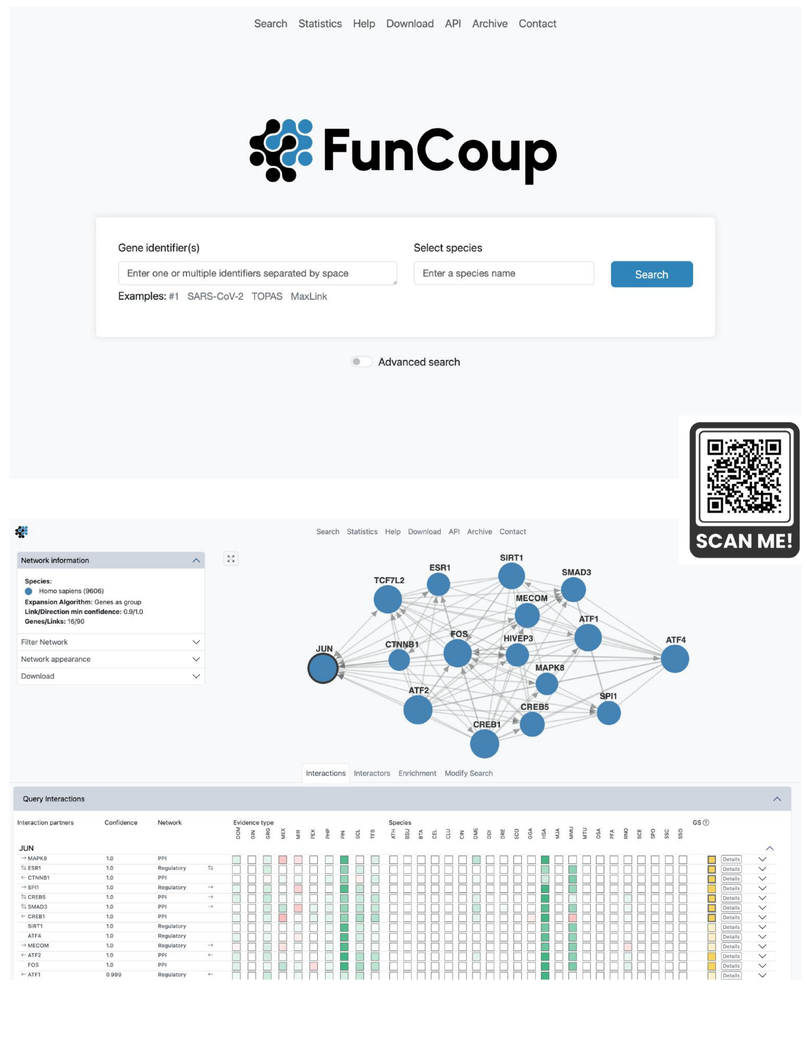
Network ORPHANET benchmark

We evaluated FunCoup 6 using ORPHANET gene sets and the Random Walk with Restart algorithm. The performance was measured by AUROC, assessing the recovery of disease-associated genes in real networks in comparison to randomized ones.

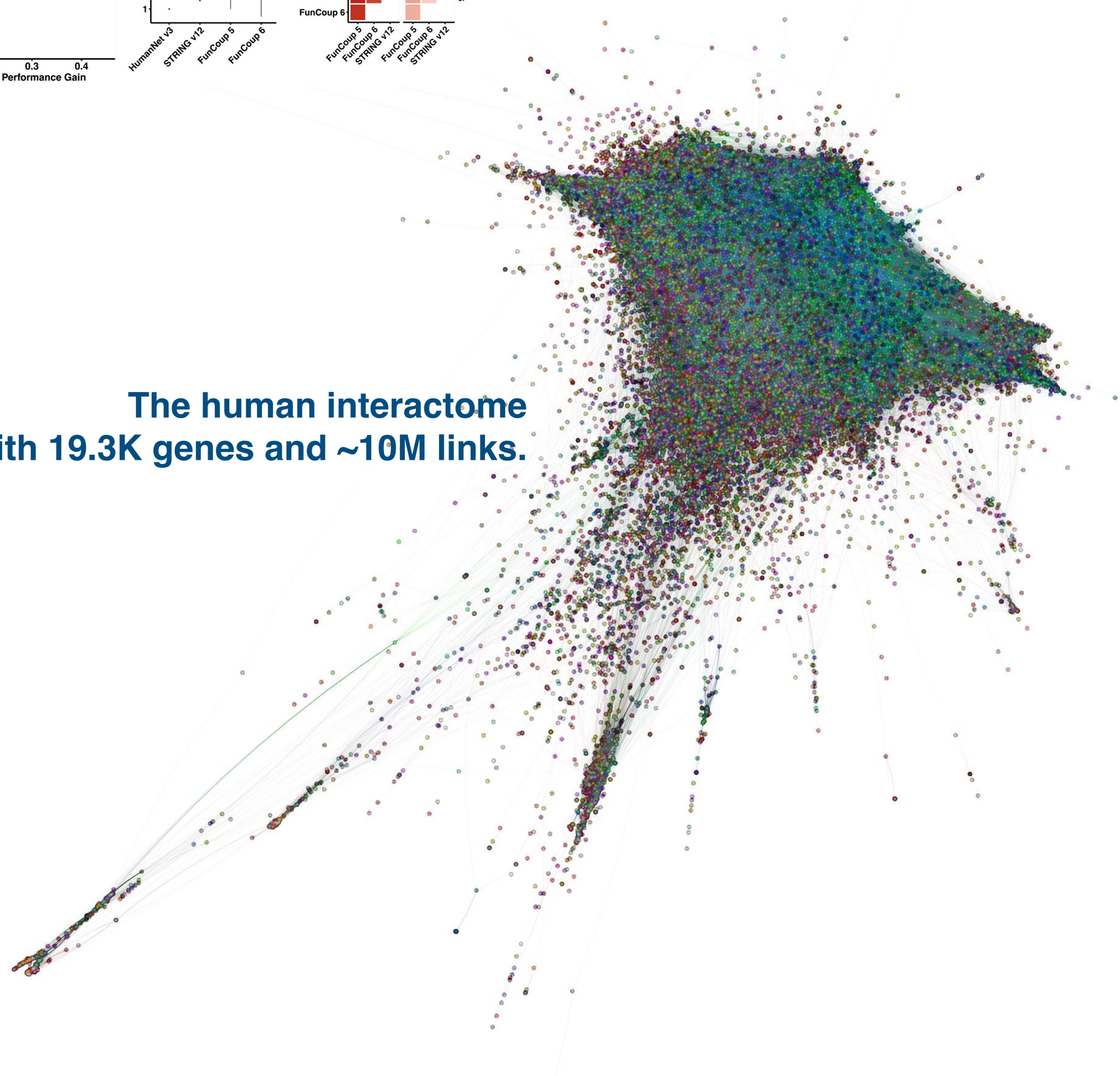


New website

The website will provide a powerful search feature which allows to retrieve the most strongly functionally coupled genes to a set of query genes. In FunCoup 6, search options will include Maxlink and TOPAS, in addition to the previous group and individual gene queries. FunCoup 6 is developed in Python (v3.10.8) using the Django framework (v4.1.1). The data is stored in a PostgreSQL database. The application is deployed in a Docker container, managed with Docker Compose (v2.17.2).



The human interactome with 19.3K genes and ~10M links.



python™ django Cytoscape graphistry v2.40.73

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